

Test Case ID	 Test Type	Test Scenario	Test Case	Pre Conditions	Test Step	Input Data	Expected Result	Actual Results	 Pass/Fail
Example	Expected	Add Item Button Functionality	Check output of item on treeview proceeded by button press	Entry widget must have a input	1. Open Window 2. Navigate to interactive widgets 3. Input a task that would like to be completed in entry 4. Press the Add Item Button	- Do the laundry	Display "Do the laundry" in the treeview widget	Displayed the "Do the laundry" task in the treeview widget	Pass
1	Expected, Boundary	Acceleration and Deceleration	Check if pressing "D" key would accelerate the player character to a maximum top speed of 10	Game Start	1. Open Window 2. Press D key	D Key	Player Character would accelerate right with a max top speed of 10	Player did accelerate to a max top speed of 10 units per second	Pass
2	Expected	Acceleration and Deceleration	Check if the corresponding directional inputs alongisde the X and Z axis are fully functioning	Player is on the ground and game is playing	1. Open Window 2. Press W or S for up and down 3. Press D or A for right and left	WASD Controls	Player would move in the corresponding direction of (up, down, left right), based on the inputs changing the velocities of the player's relative direction	Player fails to accelerate and decelerate in the right directions	Fail
3	Expected	Acceleration and Deceleration	Check if the player that is jumping falls back to the ground	Player must be elevated	1. Open Window 2. Press Spacebar	Spacebar	Player is to be pulled back to the ground at a constant rate	Player managed to fall at a constant rate back towards the ground	Pass
4	Expected	Character Directional Rotation	Check if player follows the orientation of the direction from its relative location from the cursor	Cursor on Game Window	1. Open Window 2. Move Cursor across game screen	Mouse Cursor Movement	Player's relative rotation will be facing in whatever location the mouse cursor is at by calculating a difference vector between them	Player body did not follow the orientation of the player at a constant rate relative to the coordinates of the mouse cursor to the player body giving an accurate degree angle change.	Fail
5	Expected	Test Dummy	Check if the dummy changes colour when it is hit by mouse cursor	Dummy must be on test baseplate	1. Open Window 2. Aim cursor at dummy	Mouse Cursor Movement	Dummy will change to a pinkish colour when collided with the player, and back to cyan when this condition is not further met	Dummy did not change colour due to a poor reference in material override	Fail
6	Expected	Player Gun Weapon	Check if the gun direction sticks to the player direction	Player must be visible on the game window	1. Open Window 2. Move Cursor across game screen	Mouse Cursor Movement	Gun will pivot around the player character maintaining constant distance during pivotal rotation	Gun managed to keep its orientation pivoting around the axis of the player template body	Pass
7	Expected	Player Gun Weapon	Check if the gun shoots bullets in the correct direction	Game start	1. Open Window 2. Aim Mouse Cursor 3. Left Click Input	Left Click input	Bullet when instantiated should be propelled in the direction that the player is facing	Bullets sprayed everywhere and continued to stick to the player character	Fail
8	Expected	Player Gun Weapon	Check that bullet is affected by gravity	Bullet must be present on the game window	1. Open Window 2. Aim Mouse Cursor 3. Left Click Input	Left Click input	Bullet when instantiated should immediately be affected by the rate of gravity and hit the ground	Bullet did not fall towards the ground at a constant rate	Fail

9	Expected	Player Gun Weapon	Check if bullet is destroyed upon contact of the ground or enemies	Bullet must be present on the game window	1. Open Window 2. Aim Mouse Cursor 3. Left Click Input	Left Click input	Bullet will perform queue_free function and disappear from scene	Bullet managed to perform the queue_free function and get destroyed from the bottom plate and the enemy body	Pass
10	Expected	Timers	Check if seconds decrease at a constant rate	Game start	1. Open Window	(nil)	Timer will count down from 5 minutes and proceed to decrease till it has hit 00 : 00	Timer label has constantly changed its numerially value over a constant rate	Pass
11	Expected, Boundary	Modifier Loader/Randomiser	Check if the a selection of four cards is chosen from an array to be put into a hbox	Timer Finished	1. Open Window 2. Wait for timer to finish	(nil)	Cards will be aligned in a 4 way column upon one row	Successfully aligned appropriately	Pass
12	Expected	Modifier Loader/Randomiser	Check if once row is full of four card, it will proceed to make another row for the next 4	Timer Finished	1. Open Window 2. Wait for timer to finish	(nil)	Cards will be aligned in a 2 way row by 4 way column	Successfully aligned appropriately	Pass
13	Expected, Boundary	Timers	Check if pregame timer will proceed at scene ready	Game start	1. Open Window	(nil)	Countdown from 3 seconds	Successfully performed action	Pass
14	Expected, Boundary	Timers	Check if midgame timer will proceed after pregame timer has finished	Timer Finished	1. Open Window 2. Wait for timer to finish	(nil)	Mid game timer will begin counting down from 2 minutes	Maintained a constant timer without decreasing value	Fail
15	Expected	Multiplayer Capabilities	Check if the host player is able to create a hosted server	Game start	1. Open Window 2. Press Host Button	- Host Button Pressed	Host will create a new game under being authority of the created server	Successfully created hosted local server	Pass
16	Expected	Multiplayer Capabilities	Check if the other client is able to join the hosted server	Host Server Active	1. Open Window 2x 2. Press Host Button on Window 1 3. Press Client Button on Window 2	- Host Button Pressed - Client Button Pressed	Client will join the same local game as the Host	Successfully had client join hosted server through mutual api local port	Pass
17	Expected	Multiplayer Capabilities	Check if the player is able to see the corresponding actions of the other player	Host Server Active Client Joined Server	1. Open Window 2x 2. Press Host Button on Window 1 3. Press Client Button on Window 2 4. Movement Keys (WASD)	- Host Button Pressed - Client Button Pressed - Movement Buttons Pressed	Sync player actions with other local clients in game	Successfully synced both player location and rotation between both players	Pass
18	Expected	Multiplayer Capabilities	Check if the shooting function is synced locally between players	Host Server Active	1. Open Window 2x 2. Press Host Button on Window 1 3. Press Client Button on Window 2 4. Press Left click to shoot	Left Click	Bullets will fire from respective players	Successfully displayed bullets on both character screens	Pass
19	Expected	Multiplayer Capabilities	Check if players connect to global camera	Host Server Active	1. Open Window 2x 2. Press Host Button on Window 1 3. Press Client Button on Window 2	Mouse Cursor Movement	Players will share the same global camera node	Camera directory was not properly referenced therefore game crashed	Fail
20	Expected	Player Identification Customisation	Check if local player identification has checked for player colour change	3 Window Instances opened	1. Open Window 3x 2. Press Host Button on Window 1 3. Press Client Button on Window 2 4. Press Client Button on Window 3	Left Click	Player colours will be determined by local game join status and priority	Player colour only appeared locally to each client	Fail

21	Expected	Modification Player Synchronisation	Check if player modification menu is the same displayed as the other players	3 Window Instances opened	1. Open Window 3x 2. Press Host Button on Window 1 3. Press Client Button on Window 2 4. Press Client Button on Window 3	Left Click	Players will display the same modification menu where players are able to diable one anothers card availabilities when playing	Each client independently shared different seed values	Fail
22	Expected	Modification Player Synchronisation	Check if seed is connected between between both clients	4 Window Instances Opened	1. Open Window 4x 2. Press Host Button on Window 1 3. Press Client Button on Window 2 4. Press Client Button on Window 3 5 Press Client Button on Window 4	Left Click	Players will carry the same seed between clients to have similar randomised card options to each other player	Failed to share these values among clients	Fail
23	Expected	Dummy	Check colour change for separate dummies	4 Window Instances Opened	1. Open Window 4x 2. Press Host Button on Window 1 3. Press Client Button on Window 2 4. Press Client Button on Window 3 5 Press Client Button on Window 4 6. Shoot On each client	Left Click	Client Reference of shooting an object should be shared on all clients and host	Dummy On Hit colour change visible to all clients	Pass
24	Expected	Mouse Cursor	Remove current mouse	Player Window Hosting Local Game instantiated	1. Open Window 2. Move Mouse Cursor along Viewport	Mouse Cursor Movement	Mouse Cursor should turn invisible and be replaced with another unique png	Original mouse Cursor turned invisible and replaced with unique png	Pass
25	Expected	Mouse Cursor	Check that the mouse cursor follows the players mouse location	Player Window Hosting Local Game instantiated	1. Open Window 2. Move Mouse Cursor along Viewport	Mouse Cursor Movement	Mouse png should chase the exact location of the client's mouse position	Mouse png follows mouse cursor	Pass
26	Expected	Mouse Cursor	Check that mouse cursor works in 3D space	Player Window Hosting Local Game instantiated	1. Open Window 2. Move Mouse Cursor along Viewport	Mouse Cursor Movement	Mouse cursor should follow player mouse position in 3D viewport	Mouse cursor fails to chase mouse relative location in the 3D viewport	Fail
27	Expected	Mouse Cursor	Check if mouse colour changes for different player colours corresponding to their priority	Player Window Hosting Local Game instantiated w/ client players	1. Open Window in Host 2. Open 3x Windows through client 3. Move Mouse Cursor along Viewports	Mouse Cursor Movement	Mouse cursor should correspond with the same colour as the player body upon joining game	Failed to change colour based on the client ID	Fail
28	Expected	Mouse Cursor	Check if mouse cursor changes different to two different clients active at the same time	Player Window Hosting Local Game instantiated w/ client players	1. Open Window in Host 2. Open 3x Windows through client 3. Move Mouse Cursor along Viewports	Mouse Cursor Movement	Mouse Cursor should be visible to all clients with relative colours	Failed to share local colours to other clients	Fail
29	Expected	Modification Player Scene	Check if present players show their player icon	Host x Client Server	1. Open Window in Host 2. Open 3x Window in Client	Left Click	Player icon during arena phase should be visible upon joining located at the top right of the screen	Successfully checked database for active players to show corresponding player icons at the top left of the screen	Pass
30	Expected, Invalid	Modification Player Scene	Check if player that are not in game do not show	Host x Client Server	1. Open Window in Host 2. Open 3x Window in Client 3. Wait for arena timer to end	Left Click	Player icons that are not currently in the game should not be visible to other clients since they are non-existent in the current game	Check if player icons during modification stage are invisible upon invalid load	Pass

31	Expected	Modification Player Scene	Check if cards disable when clicked on	Modifier Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Wait for arena timer to end 4. Click on a card for each client	Left Click	Modification cards should turn semi transparent when clicked upon to resemble that they are disabled to the player	Successfully disabled cards when clicked on by players	Pass
32	Expected	Modification Player Scene	Check disabled feature is active amongst all clients	Modifier Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Wait for arena timer to end 4. Click on a card for each client	Left Click	Relay these disabled property to all clients upon vanity changes	Successfully broadcasted disability to other clients	Pass
33	Expected	Modification Player Scene	Check if player is properly referenced	Modifier Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Wait for arena timer to end 4. Click on a card for each client	Left Click	Properly test reference on which client authority had pressed the card most recently and prepare to provide stat changes upon priority	Failed to reference player who pressed the modification card	Fail
34	Expected	Modification Player Scene	Check if player gains move speed increase	Modifier Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Wait for arena timer to end 4. Click on a card for each client	Left Click	Player independent value will change causes the mutliplied velocity to increase to a constant of movement speed	Successfully changed value	Pass
35	Expected	Player Shooting	Check if cooldown of 1.5 for each bullet fire	Modifier Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Wait for arena timer to end 4. Click on a card for each client	Left Click	Create a temporary wait timer that increases the intervals of bullet fire between 1.5 seconds	Successfully changed value	Pass
36	Expected	Modification Player Scene	Check if player fire rate is increased	Modifier Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Wait for arena timer to end 4. Click on a card for each client	Left Click	Once the player has chosen an attack speed modifier, their fire rate should increase	Failed to modify fire rate constant	Fail
37	Expected	Gameplay Loop	Check if modifiers scene loads game hud scene	Modifier Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Wait for arena timer to end 4. Click on a card for each client (x4)	Left Click	Game Scene should traverse to the game arena scene once players have completed their selection in modifiers	Successfully loaded playable arena scene when players have completed their modifier choice	Pass
38	Invalid, Expected	Gameplay Loop	Check if players that already have a modifier selected cannot choose another	Modifier Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Wait for arena timer to end 4. Click on a card for each client	Left Click	Once a player has chosen a card their indivual available card opportunity value decreases to 0 preventing further card selection	Players successfully limited to only one card per round	Pass
39	Expected	Player Scoring	Check once player hits another with its bullet, the shooter will gain a point	Arena Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Shoot another client player	Left Click	The player once hitting another player will individually add 1 to their score in their local identity	Successfully increased point based on player reference of bullet being shot	Pass
40	Expected	Player Multiplayer Desync	Check if players will recieve the same card modifier display	Modifier Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Wait for arena timer to end	Left Click	Once players have loaded the card modifier instantiation scene, they will all recieve the same display	Players did not all recieve visible card instantiation	Fail

41	Expected, Boundary	Scoreboard	Check if score labels modify upon player score increase	Arena Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Shoot another client player	Left Click	Player score label should increase relative to their individual player listings of their player scores for other clients to see	Failed to change value corresponding to local client	Fail
42	Invalid	Scoreboard	Check if score correctly displays increments of 1 per kill and not 2x	Arena Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Shoot another client player	Left Click	Players will not provide a feedback loop where the player locally and globally adds to the players score desyncing their score numbers	Successfully prevented increase in increments from improper client x host authority	Pass
43	Expected	Scoreboard	Check if clients recieve change in score based on server	Arena Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Shoot another client player	Left Click	At the top right of the client screen the numerical score label will change dependent on the server towards the clients	Successfully changed score on all clients	Pass
44	Expected	Player Respawn	Check if players respawn at 4 randomised points on the map	Arena Stage	1. Open Window in Host 2. Open 3x Window in Client 3. Press Start Button with atleast 2 players	Left Click	Players should spawn at points relative to 4 choices for players to instantiate upon rather than being spawned at the middle of the arena for lack of strategic maneuvers	SUccessfully randomised player locations	Pass